Exact Step Counts and an Exact Halting Time for the BB(6) Champion, Proved in Coq

Vishnu Vaidyanathan
vishnu@edgi.in
ORCID 0009-0001-6244-5262
https://github.com/VishnuVaidyanathan/BB5-decimal-macrogrammar

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Abstract

The current 6-state Busy Beaver champion, found by mxdys in 2025, runs for more than $2 \uparrow\uparrow\uparrow 5$ steps. mxdys proved in Coq that it halts, using a small set of rules of the form "this tape shape reaches that tape shape". Those rules carry no step counts, and the halting time is known only as an estimate of its size. We add the missing step counts. For every move of the machine (the tick of its binary counter, a full countdown, the start of an epoch, the restart, and the final move) we give the exact number of steps as a short formula, valid for all values of the parameters. From these we build an exact expression for the halting time and for the number of ones on the final tape. We prove in Coq, with no axioms, that the machine started on the blank tape halts after exactly that many steps and not before, and that the final tape holds exactly that many ones. The numbers cannot be written out, but the theorems fix which numbers they are. The halting time is close to two thirds of the square of the number of ones. The moves were first found by reading the transcript of the machine, the method of our earlier paper on the BB(5) champion.

1 Introduction

The machine studied here is

`1RB1RA_1RC1RZ_1LD0RF_1RA0LE_0LD1RC_1RA0RE`,

found by mxdys in June 2025. It is the current BB(6) champion: it halts, and its running time and its number of ones lie between $2 \uparrow\uparrow 2 \uparrow\uparrow 2 \uparrow\uparrow 10$ and $2 \uparrow\uparrow 2 \uparrow\uparrow 2 \uparrow\uparrow 11$, which is more than $2 \uparrow\uparrow\uparrow 5$ [2]. The proof that it halts is a Coq development by mxdys [1]. That proof describes the tape by a few numbers and shows how these numbers change. Every statement in it has the form $c \to^* c'$: the configuration c reaches c' after *some* number of steps. The number of steps is never stated, and so the proof does not say when the machine halts.

This paper supplies the step counts.

What is not new. The tape shapes, the rules that take one shape to the next, the fact that the machine counts in binary, the size of the running time and the proof of halting are all the work of mxdys [1, 2]. We found the same moves independently by reading the transcript of the machine, but they are the same moves.

Contributions.

1. An exact step count for each move of the machine, for all values of its parameters (Theorems 1–5). Each is a linear formula, except the full countdown, which is a quadratic in 2^b.

2. An exact expression for the halting time, and a Coq proof that the machine halts after exactly that many steps and at no earlier step (Theorem 7).

3. An exact expression for the number of ones on the final tape, also proved in Coq (Theorem 8).

4. The transcript words of each move (Section 8). These are tested against simulation but are not yet stated in Coq.

The Coq file is self-contained (about 890 lines, Coq 8.18, no axioms). It does not import the development of mxdys; it proves the moves again in its own small model, now with step counts.

2 Preliminaries

The machine. The transition table is

	A	B	C	D	E	F
read 0	1RB	1RC	1LD	1RA	0LD	1RA
read 1	1RA	1RZ	0RF	0LE	1RC	0RE

The machine starts in state A on a blank tape. The only halting pair is B reading 1. We count the halting transition as a step, as in [3].

Configurations. We write $0^\infty\, u\, [\mathrm{s}\ x]\, v\, 0^\infty$ for the tape $u\,x\,v$ surrounded by blanks, with the head in state s on the cell holding x. Powers denote repetition.

Transcripts. The transcript of a run is the list of state–symbol pairs it visits [4]. The machine has twelve pairs, so we write each as one character, $d(\mathrm{s}, x) = 2 \cdot \mathrm{idx}(\mathrm{s}) + x$, with the values ten and eleven written `a` and `b`: A0 = `0`, A1 = `1`, ..., E1 = `9`, F0 = `a`, F1 = `b`. The halting pair B1 is the digit `3`.

Three basic words. Almost every step of the run belongs to one of three repeated words.

- `1` (A1): the head walks right over a one.
- `479` (C0, D1, E1): a 0 under the head moves one cell to the left through a block of ones: $1\,1\,[\mathrm{C}\ 0] \to 1\,[\mathrm{C}\ 0]\,1$.
- `5b9` (C1, F1, E1): the block 111 becomes 001 and the head moves three cells to the right.

3 Tape shapes

The shapes below are those of mxdys [1], written for this machine.

Which run. In [1] the transition table of the champion appears, with the states renamed, as two modules: `TM62`, started in the champion's state D, and `TM63`, started in its state B. The champion itself starts in state A. Its configuration after 15 steps is the configuration of `TM63` after 9 steps, and the two runs are the same from there on (`Eqv_v2.TM872` in [1]). So `TM63` is the champion, six steps behind, and its halting proof is the halting proof of the champion. `TM62` is a different run that obeys the same rules, with a slightly different left end of the tape. Everything in this paper is about the run from state A on the blank tape.

Counters. A counter is a number written in binary, high bit first, with the bit 1 written as the block 100 and the bit 0 written as the block 111. Counters count down.

The inner shape S_0. A counter with any tape X to its left, the head in state E on the last cell of the counter, and to the right the block

$$R(b) \;=\; 0\ 11100\ (111)^{1+b}\ 0^{\infty}.$$

The outer shape S_1. The whole tape at the start of an epoch is

$$S_1(\beta, q, b) \;=\; 0^{\infty}\ 111011\ \ \beta\ \ 100\ \ 10\ \ 1^{q-1}\,[\mathrm{E}\ 1]\ \ R(b),$$

where β is a counter, called the *outer counter*, and q is the number of ones after the marker 10. (For $q = 0$ the head is on the 0 of the marker.) In the numbers of mxdys, $S_1(len_0, a_0, m, a, 0, b)$, the outer counter has len_0 bits and value a_0, and

$$q = m + 3a, \qquad m \in \{0, 1, 2\}.$$

So one number q carries both the marker $m = q \bmod 3$ and the length a of the inner counter, which is all zeros at the start of an epoch. This makes the rules shorter.

Throughout, $N_k = 2^k - 1$.

4 The tick

Theorem 1 (Tick; `tick`)**.** *Let a counter end in a bit* 1 *followed by $j \geq 0$ bits* 0*. From the shape S_0 with this counter and right block $R(b)$, the machine reaches the shape S_0 in which that bit* 1 *has become* 0*, the j bits after it have become* 1*, and the right block is $R(b+1)$. It takes exactly*

$$12b + 12j + 48 \quad \textit{steps.}$$

In other words the counter goes down by one, the right block gains one 111, and j is the length of the borrow. This is the lemma `Inc0` of [1], with the step count added.

Theorem 2 (Countdown; `full_ticks`, `T_closed`)**.** *Let the lowest k bits of a counter all be* 1*. From S_0 with right block $R(b)$ the machine reaches S_0 with those k bits all* 0 *and right block $R(b + N_k)$, in exactly $T(k, b)$ steps, where*

$$T(k, b) \;=\; 6N_k^2 + 54N_k + 12\,b\,N_k - 12k.$$

Idea of the proof. To clear $k+1$ bits, clear the low k bits, do one tick with borrow k, and clear the low k bits again. This gives the recursion $T(k+1, b) = T(k, b) + \big(12(b + N_k) + 12k + 48\big) + T(k,\, b + N_k + 1)$ with $T(0, b) = 0$, and the closed form follows by induction on k. No sum over the borrow lengths is needed. □

5 The epoch

An epoch starts at a shape S_1 whose outer counter is not zero.

Theorem 3 (Epoch; `load`, `epoch`)**.** *Let the outer counter β end in a bit* 1 *followed by $j_0 \geq 0$ bits* 0*, and let β' be β minus one. Then*

$$S_1(\beta, q, b) \;\longrightarrow\; S_1\big(\beta',\ q + 3b + 5,\ 2^b - 1\big)$$

in exactly

$$\underbrace{4q + 12j_0 + 3b + 63}_{\textit{load}} \;+\; \underbrace{T(b, 0)}_{2^b - 1\ \textit{ticks}} \quad \textit{steps, where } T(b, 0) = 6N_b^2 + 54N_b - 12b.$$

The load moves the right block $(111)^{1+b}$ into a new inner counter as the value $2^b - 1$, and takes one off the outer counter. The ticks then count the inner counter down to zero, which builds a right block of $2^b - 1$ units. These are the lemmas `Rst2`, `Rst1`, `Rst0` of [1] (called Inc2, Inc1, Inc0 on the wiki [2]). In their numbers the load costs $12a + 3b + 12j_0 + 63 + 4m$ steps. Since q grows by $3b + 5$, the marker $m = q \bmod 3$ goes $2 \to 1 \to 0 \to 2$ from one epoch to the next.

6 Restart and halt

When the outer counter is zero, the head runs to the left end of the tape. What happens next depends on q mod 3.

Theorem 4 (Restart; `restart`)**.** *Let the outer counter be L zero bits and let $q = 3a + 1$. Then*

$$S_1(0^L,\ 3a + 1,\ b) \longrightarrow S_1(\beta^*,\ 3b + 2,\ 2^b - 1)$$

in exactly $12L + 12a + 3b + 97 + T(b, 0)$ steps, where the new outer counter β^* has $L + a + 2$ bits: $L + 1$ ones, then a zero, then a ones. Its value is $2^{L+a+2} - 2^a - 1$.

Theorem 5 (Halt; `last_move_cfg`, `halt_ones`)**.** *Let the outer counter be L zero bits and let $q = 3a$. From $S_1(0^L, 3a, b)$ the machine halts after exactly*

$$12L + 12a + 60 \quad \textit{steps,}$$

and the final tape holds exactly $L + a + 3b + 16$ ones.

The count of ones can be read off the final tape: 6 in the left end 1110111, $L + 4$ in what was the outer counter, $a + 1$ in the middle block, 2 written just before the halt, and $3b + 3$ in the right block, which is not touched.

7 The whole run

Proposition 6 (`init`)**.** *From the blank tape the machine reaches $S_1(111,\ 20,\ 63)$ after exactly* 27,965 *steps. In the numbers of mxdys this is $S_1(3, 7, 2, 6, 0, 63)$.*

By Theorem 3 the first epoch then takes $332 + T(63, 0)$ steps. It ends at step

$$510{,}423{,}550{,}381{,}407{,}695{,}582{,}443{,}536{,}695{,}552{,}928{,}613 \approx 5.1 \times 10^{38}$$

(`first_epoch`). The second epoch has $2^{2^{63}-1} - 1$ ticks, so no later step number can be written out.

Many epochs. Between restarts only two numbers change, q and b, together with the outer counter. One epoch is the map $(q, b) \mapsto (q + 3b + 5,\ 2^b - 1)$ with the cost of Theorem 3. The Coq file defines two functions that add up the costs of a long run of epochs by recursion on the shape of the outer counter, in the same way as the proof of Theorem 2: `runF` k clears k low bits that are all 1 ($2^k - 1$ epochs), and `runTop` n z clears n ones that sit above z zeros. The theorems `runF_ok` and `runTop_ok` say that the machine follows these functions exactly.

Restart or halt. Two simple facts decide the whole run (`runF_inv`, `runTop_inv`): b is always odd, and therefore each epoch adds 2 to q modulo 6. Together with $N_k \equiv 0$ or $1 \pmod 3$ for k even or odd, this gives the three passes found by mxdys:

Pass	Outer counter at the start	Epochs mod 3	q at the end
1	111 (value 7)	1	$6t+4 = 3a_1+1$, a_1 odd: restart
2	$2^{a_1+5} - 2^{a_1} - 1$	1	$6t+7 = 3a_2+1$, a_2 even: restart
3	$2^{a_1+a_2+7} - 2^{a_2} - 1$	2	$6t+9 = 3a_3$: halt

Theorem 7 (Exact halting time; `bb6_halts_exactly`)**.** *Let H be the sum of:* 27,965*; the costs of the* 7 *epochs of pass 1; the first restart; the costs of all epochs of pass 2; the second restart; the costs of all epochs of pass 3; and the final move* $12L_3 + 12a_3 + 60$*, where* $L_3 = a_1 + a_2 + 7$*. Then the machine, started on the blank tape, is halted after H steps and is not halted after any smaller number of steps.*

In Coq, H is the closed term `halting_time`, built from `runF`, `runTop`, T and the formulas of Theorems 3–5. Coq never evaluates it.

Theorem 8 (Final ones; `bb6_final_ones`)**.** *The final tape holds exactly* $L_3 + a_3 + 3b_3 + 16$ *ones, where b_3 is the value of b at the last shape.*

Since b_3 is far larger than L_3 and a_3, the number of ones is very nearly $3b_3$. This agrees in size with the estimates on the wiki [2].

Runtime against ones. Let b' be the value of b at the start of the last epoch, so that $b_3 = 2^{b'} - 1$. By Theorem 3 the ticks of that one epoch take $6b_3^2 + 54b_3 - 12b'$ steps. So, writing σ for the number of ones,

$$H \;=\; 6b_3^2 + 54b_3 + R, \qquad \sigma \;=\; 3b_3 + r, \qquad r = L_3 + a_3 + 16,$$

where R collects every other step of the run. Both R and r are tiny beside b_3: each earlier epoch is smaller than the next by a whole exponential, so all of them together cost about $(\log b_3)^2$ steps, and L_3 and a_3 are about $\log b_3$ or less. Hence

$$H \;\approx\; \tfrac{2}{3}\,\sigma^2.$$

The two displayed identities follow from the theorems above. The size of R and r is a pen-and-paper estimate and is not stated in Coq. As a small test, a starting shape whose run ends with $b = 127$ gives $H/\sigma^2 = 0.61$ and $R = 1044$. A running time of the order of σ^2 is what one expects of a machine that sweeps back and forth over its tape.

8 Transcript words

The moves above were first seen in the transcript. Write $\Omega = \texttt{479}$ and $\Lambda = \texttt{5b9}$. The transcript of each move is one word with a few counts:

Move	Word
Tick, $j \geq 1$	$\texttt{9}\,\Omega^{3j-1}\,\texttt{478602}\,\Lambda^{j+1}\,\texttt{4786025a}\,\texttt{1}^{3b+3}\,\texttt{02}\,\Omega^{3b+5}\,\texttt{47879}\,\Omega^2\,\texttt{47}$
Tick, $j = 0$	$\texttt{8602}\,\Lambda\,\texttt{4786025a}\,\texttt{1}^{3b+3}\,\texttt{02}\,\Omega^{3b+5}\,\texttt{47879}\,\Omega^2\,\texttt{47}$
Load	$\texttt{9}\,\Omega^{q-1}\,\texttt{47878602}\,\Omega^{3j_0+3}\,\texttt{478602}\,\Lambda^{j_0+1}\,\texttt{5a}\,\texttt{1}^{q+3}\,\texttt{02}\,\Lambda^{b+1}\,\Phi$
Restart	$\texttt{9}\,\Omega^{q-1}\,\texttt{47878602}\,\Pi\,\Lambda^{a+1}\,\texttt{5b8602}\,\Lambda^{b+1}\,\Phi$
Halt	$\texttt{9}\,\Omega^{q-1}\,\texttt{47878602}\,\Pi\,\Lambda^{a+1}\,\texttt{5a03}$

Here

$$\Phi = \texttt{4786025a02}\,\Omega^2\,\texttt{47879}\,\Omega^2\,\texttt{47},$$
$$\Pi = \Omega^{3L+5}\,\texttt{47879}\,\Omega\,\texttt{4786025b8602}\,\Lambda^{L+2}\,\texttt{478602}.$$

The word Φ closes the load and the restart, and Π is the turn at the left end of the tape, and for $q = 0$ the opening $\texttt{9}\,\Omega^{q-1}\,\texttt{47878602}$ is replaced by `878602`. The lengths of these words are the step counts of the theorems. The halt word ends in `3`, the only occurrence of the halting pair in the whole run.

The words were checked against direct simulation on 228 cases with small parameters (script `check_words.py` in the repository). The Coq proofs follow exactly these words, one lemma for each repeated word and a computation for each fixed piece, but the digit strings are not yet stated as theorems.

Remark 9. In the first 30,000,000 steps from the blank tape, the digit `a` cuts the transcript into 2,305 pieces. Of these, 2,299 are exactly one tick word each (cut at a different point of the word), with the counts given above. The other six lie in the first 800 steps and around the first load. This is how the tick was found, before the proof of mxdys was read.

9 The Coq proof

The development is the file `bb6/BB6Steps.v` in the repository (about 890 lines, Coq 8.18). `Print Assumptions` reports that `bb6_halts_exactly`, `bb6_final_ones`, `first_epoch` and `T_closed` are closed under the global context: no axioms are used.

Model. As in [3], a configuration is a state, the list of cells to the left of the head (nearest first), the cell under the head, and the list of cells to the right. An empty list stands for blanks. `steps n c` applies the transition table n times.

Structure. Three lemmas handle the repeated words `1`, `479` and `5b9` for any number of repeats, by induction. Each move is then a chain of these lemmas and of short fixed computations, with the step counts added along the chain. Counters are lists of bits, so that a countdown is an induction on the list. The numbers a_1, a_2, a_3 are tower-sized, but they only ever appear as variables.

The model is the right machine. The start of the run is checked by computation inside Coq: 821 steps from the blank tape give a shape S_0, and Theorem 2 then gives Proposition 6.

Independent test. A Python script in the repository, `check_steps.py`, starts the machine from 132 small shapes S_1 that halt, and predicts the halting step and the final ones using only the formulas of this paper and the rules of mxdys. All 132 agree with plain simulation.

What is not in Coq. The transcript words of Section 8. Also, this proof and that of mxdys use different tape models and are not linked to each other; they agree on the shapes and on the three passes.

10 Discussion

What kind of result this is. Nothing here changes what is known about the behaviour of the machine. The result is exact bookkeeping on top of a known structure: the proof of halting said that the machine stops, and this paper says after how many steps, as a definite expression.

We do not know of an earlier exact expression for the halting time of this machine, but one may exist in unpublished notes or code of the bbchallenge community.

The step counts are simple. Every move costs a linear number of steps in its parameters, and a countdown costs a quadratic in 2^b. The enormous running time comes only from the rule $b \mapsto 2^b - 1$, applied once per epoch.

Transcripts as a way to find rules. In [3] the transcript of the BB(5) champion was described by a grammar of repeated words with growing counts. The same reading of the BB(6) champion's transcript gave the tick, its two counts and its cost before any tape analysis was used. For the later moves this is not possible: the first epoch alone takes about 5×10^{38} steps, so no simulation reaches a restart. There the rules of mxdys were needed to know where to look, and small starting shapes were used to measure the costs.

Future work. (1) State the transcript words in Coq. (2) Derive small facts about the halting time, such as its remainder modulo a small number. (3) Apply the same counting to the other machines covered by [1].

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References

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